



Edge-AI based multi-criteria optimization framework for dynamic EV charging with real-time grid load, traffic, and user behavior integration

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Received: 21 April 2025 / Accepted: 21 July 2025 / Published online: 13 August 2025

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Abstract

As the transition toward electrified transportation gains momentum, the challenge of managing electric vehicle (EV) charging in real-time—while minimizing energy costs and maintaining grid stability—has become increasingly critical. Existing centralized control methods often struggle to adapt to the rapid influx of heterogeneous data, including traffic conditions, dynamic grid status, and user-specific preferences. To address this, we introduce an Edge-AI based optimization framework that manages EV charging by simultaneously considering multiple criteria in real-time. Our system synthesizes live data from traffic networks, electric utilities, and user mobility patterns to deliver context awareness charging recommendations. The framework employs a hybrid algorithm combining Reinforcement Learning with the Analytic Hierarchy Process (AHP) to evaluate and prioritize conflicting objectives such as reduced wait time, minimized grid strain, lower electricity costs, and decreased emissions. By shifting decision-making closer to the source through edge computing, the model achieves low-latency responsiveness and enhanced resilience against connectivity failures. We tested the framework using a combination of real-world and synthetic datasets to simulate diverse urban conditions. Results indicate up to 23.5% shorter wait times, 18.2% improvement in grid utilization, and 21.7% cost reduction compared to conventional approaches. These findings suggest that edge-intelligent, multi-objective systems can play a pivotal role in developing adaptive, scalable EV charging infrastructures, especially in alignment with U.S. smart grid and sustainability initiatives.

Keywords Edge AI · Electric vehicle charging · Multi-criteria optimization · Grid load forecasting · Smart mobility · Real-time traffic integration

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Mathematics Subject Classification (2000) 68T05 · 90C29 · 68W25 · 90C90 · 68M10

1 Introduction

The rapid evolution of electric mobility, driven by advancements in battery systems and sustainability mandates, is transforming transportation systems worldwide. In nations such as the United States, legislative and industry-backed initiatives are accelerating the adoption of electric vehicles (EVs) as a means to reduce emissions and improve energy efficiency. Projections by the U.S. Department of Energy suggest that by the end of this decade, EVs could comprise over a third of new vehicle sales, necessitating a proportional upgrade to the underlying charging infrastructure. However, this shift is accompanied by growing concerns the reliability, responsiveness, and efficiency of existing charging systems.

While the number of charging stations is increasing, current solutions often fall short in adapting to the dynamic conditions found in urban and semi-urban environments. Typical charging systems rely on static logic or centralized scheduling protocols that are unable to respond effectively to real-time variations in traffic patterns, energy pricing, grid load, and driver behavior. This inflexibility can result in suboptimal station usage, extended wait times, bottlenecks during peak hours, and even instability in local power networks due to uncontrolled load surges.

Artificial intelligence (AI) offers a pathway toward smarter, adaptive charging solutions. Machine learning models, and more recently, reinforcement learning techniques, have been explored to support decision-making in charging systems. These approaches, however, are predominantly implemented through cloud-based infrastructures, which may struggle with latency, bandwidth limitations, or fail to operate under intermittent network conditions. Moreover, many of these AI-driven systems are designed around optimizing a singular outcome—such as minimizing time or cost—while neglecting the multifactorial nature of real-world EV charging that demands a balance between user satisfaction, grid impact, financial cost, and environmental goals.

To address these limitations, a new direction is needed—one that supports multi-objective optimization and enables localized decision-making. This paper explores such an approach by proposing a framework based on Edge Artificial Intelligence (Edge-AI), where intelligent algorithms are embedded at the station or vehicle level to deliver rapid and autonomous responses. Unlike cloud models, edge computing places computational processes closer to where data is generated, enhancing the system's ability to function in time-sensitive scenarios with minimal dependence on remote servers.

Another shortcoming of current models lies in the fragmented use of data. While individual datasets—such as vehicle telemetry or grid status—have been used to inform decisions, few systems integrate multiple sources such as urban traffic conditions, dynamic electricity tariffs, real-time grid strain indicators, and personalized driver behavior. For instance, traffic congestion may influence not just route planning but also station selection, as delays may lead to queue formation at certain loca-

tions. Similarly, time-of-use pricing from utility companies could incentivize shifting charging sessions to periods of lower demand or higher renewable energy availability.

This paper proposes a context awareness optimization framework that uses Edge-AI to synthesize and act upon real-time information from diverse domains. Our system incorporates data on road conditions, utility load profiles, vehicle state-of-charge, and driver-specific patterns to generate customized charging recommendations. At its core is a hybrid algorithm that combines Reinforcement Learning (RL)—which dynamically adapts to environmental feedback—and the Analytic Hierarchy Process (AHP)—a structured technique for evaluating multiple, often conflicting objectives. This combination allows for the prioritization of competing goals such as minimizing queue duration, reducing grid load, lowering electricity cost, and improving environmental sustainability.

By processing this data at the network's edge, our model achieves a high degree of responsiveness and autonomy, which is particularly important for mission-critical scenarios and areas with weak or unreliable internet access. The system architecture also supports scalability, allowing multiple stations and vehicles to make coordinated decisions without requiring a centralized controller.

This multi-criteria optimization approach offers several advantages. First, it enables personalized, real-time charging strategies that account for user priorities—such as urgency of travel, willingness to pay, or environmental preferences. Second, it helps distribute charging demand across a network of stations in a way that smooths peak usage and mitigates strain on the electrical grid. Third, it allows stakeholders—utilities, city planners, and infrastructure providers—to incorporate sustainability metrics like carbon intensity and renewable generation availability directly into operational strategies.

The importance of such a solution is further underscored by recent developments in U.S. energy policy. Federal programs, including those under the Bipartisan Infrastructure Law and Inflation Reduction Act, have committed billions to modernize the grid and expand the EV ecosystem. These policies emphasize not only quantity—more stations, more vehicles—but also quality, including smarter, cleaner, and more resilient transportation energy systems. Our framework aligns with these priorities by delivering a technology solution that is adaptive, decentralized, and responsive to real-time conditions.

This study contributes to the academic and engineering communities in several key ways. First, it presents an integrated edge-computing framework that combines multiple optimization methods and data types into a unified system for EV charging. Second, it offers a flexible and scalable deployment model suitable for real-world environments, including urban, suburban, and highway scenarios. Third, the proposed method is validated through simulations using real and synthetic data, providing quantitative evidence of performance improvements in operational efficiency and energy usage.

In the following sections, we review related work on AI-based EV charging strategies and identify unresolved challenges. We then present the design and mathematical foundations of our proposed framework, including system architecture, optimization models, and learning algorithms. Experimental results from our performance evaluation are discussed in depth, followed by a technical analysis of system behavior under

varying conditions. Finally, we summarize our findings and propose directions for future development.

2 Related work

2.1 Decentralized processing through edge computing

The growth of electric vehicle (EV) usage has driven interest in computational frameworks that can support increasingly complex and decentralized energy management tasks. One solution that has gained traction is edge computing, which enables processing data at or near the source rather than relying solely on centralized cloud servers. Al Gafri et al. [1] explored edge-based charging infrastructure and highlighted improved real-time response and autonomy in decision-making at the station level. In a more extensive assessment, Arévalo et al. [2] analyzed the application of edge devices across electric transportation systems and noted the potential for enhanced latency control and resilience, particularly in volatile grid conditions.

However, many edge-oriented systems are still rudimentary, often serving only as localized data filters or basic execution nodes. Most lack sophisticated predictive capabilities or the capacity to adapt to fast-changing traffic and grid variables. Without deeper integration of intelligent algorithms, edge systems remain underutilized in dynamic urban energy environments. Recent work by Alghamdi et al. [3] has also emphasized the integration of *decentralized energy storage* for managing EVs' charging and discharging in smart cities. This decentralized approach supports real-time load balancing, crucial for managing energy demand fluctuations in urban areas.

2.2 Adaptive learning models for EV charging decisions

Artificial intelligence (AI), especially reinforcement learning (RL), has become central in optimizing energy flows within EV charging networks. These learning-based models are especially suited for problems characterized by feedback loops and state changes, such as those present in dynamic transportation systems. RL models have been shown to outperform static or rule-based schedulers by allowing systems to adapt to evolving demand and infrastructure constraints. Zhao [4], for instance, employed deep reinforcement learning within a Markov Decision Process (MDP) framework to regulate charging behaviors in response to congestion, availability, and vehicle movement patterns.

Expanding on this, several researchers have begun to explore multi-agent reinforcement learning for charging coordination, particularly where station and vehicle decisions are interdependent. In a recent study, Mosammam et al. [5] implemented Q-learning to manage the trade-off between energy efficiency and battery health in plug-in hybrid electric vehicles. Yet, even these advanced learning systems often fall short in addressing multiple optimization targets concurrently, making them less applicable for holistic system-level control.

This direction aligns with earlier work by Ahmed et al. [6], who investigated cooperative spectrum sensing in cognitive radio vehicular networks, emphasizing

the potential of distributed agents to make collective decisions in dynamic and bandwidth-constrained environments—an architectural concept that is highly applicable to edge-based EV charging coordination frameworks.

In a broader context, Said et al. [7] proposed *advanced scheduling protocols* for home charging with time-of-use pricing, illustrating how charging strategies can be adapted to dynamically changing electricity prices. This approach provides insights into how adaptive learning models can incorporate pricing strategies into EV charging decisions.

2.3 Decision models based on multiple objectives

In practical deployment scenarios, optimizing EV charging infrastructure often involves trade-offs that are inherently multi-dimensional—such as balancing economic, technical, and environmental objectives. For this reason, a growing number of researchers have adopted Multi-Criteria Decision-Making (MCDM) methods, including tools like the Analytic Hierarchy Process (AHP), fuzzy logic, and composite scoring models.

Al Gafri et al. [8] applied AHP in selecting locations for EV charging stations based on metrics such as cost, accessibility, and grid readiness. Their results indicated that structured decision hierarchies can enhance transparency and reproducibility in siting decisions. Similarly, Kumar et al. [9] combined AHP with a differential search optimization algorithm to create a strategy for sustainable EV infrastructure planning, emphasizing adaptability to both energy policy and user behavior.

Yet, many of these frameworks are implemented in static or simulation-only contexts. They rarely operate in real-time or ingest live data, which limits their practical value in urban ecosystems that evolve hour by hour. Furthermore, Said and Mouftah [10] introduced a *load management protocol* for EVs charging and discharging, further integrating AHP into a real-time decision-making process for dynamic load balancing and efficient energy distribution.

2.4 Real-time data fusion for charging optimization

One of the most underutilized opportunities in EV infrastructure optimization is the integration of diverse real-time data sources—from road congestion analytics to user travel histories and fluctuating electricity prices. While numerous studies consider these factors individually, comprehensive data fusion remains uncommon. Suanpang and Jamjuntr [11] noted that while AI is increasingly applied in EV energy systems, integration of mobility, environmental, and grid-layer inputs into a singular model is rare.

Radenkovic and Huynh [12] focused on opportunistic energy balancing within the fog and vehicular edge networks, integrating energy state and traffic data to improve system robustness. While promising, such approaches still fall short of fully synchronizing user-specific mobility profiles, real-time traffic density, and live grid load conditions to inform dynamic optimization routines. Alghamdi et al. [13] proposed *profit maximization for EVSEs* based on renewable energy sources, which further ties in multi-objective optimization with real-time data from renewable energy systems and

EV charging stations. This work complements your proposed approach by exploring how profit maximization and sustainability can co-exist in real-time optimization.

This presents a key research gap: the need for models that combine traffic behavior, grid intelligence, and user history into a cohesive control mechanism that runs at the edge. Research by Khabbouchi et al. [14] highlights *machine learning models* for energy management, which could be useful in integrating real-time traffic, grid load, and user behavior data into a unified optimization framework for EV charging.

2.5 Identified limitations in state-of-the-art models

Despite progress in AI and computational infrastructure, the current landscape of EV charging optimization systems exhibits several shortcomings:

1. **Narrow Objective Focus:** The majority of optimization strategies address isolated goals—commonly either time minimization or energy costs reduction—without accounting for grid sustainability, carbon emissions, or user satisfaction in a unified framework [4].
2. **Overreliance on Cloud Computing:** While RL has demonstrated efficiency gains, most implementations still depend heavily on cloud-hosted environments, which are prone to latency and require stable connectivity [5].
3. **Insufficient Personalization:** Many systems overlook user diversity in urgency, energy needs, and behavioral preferences. Such oversights limit their capacity to deliver adaptive, user-centric services [12].
4. **Opaque Decision Logic:** A significant portion of current AI models offer minimal explainability, complicating validation, public deployment, and regulatory acceptance [15].

There is an urgent need for modular, explainable, and edge-deployable frameworks that not only accommodate diverse data inputs but also prioritize transparency, real-time decision-making, and resilience. The present research addresses this gap by combining Reinforcement Learning and AHP into a real-time optimization engine designed for distributed execution at EV stations or on-vehicle platforms.

3 Methods

3.1 System overview and problem scope

The proposed system aims to optimize electric vehicle (EV) charging using a hybrid intelligence architecture, combining Reinforcement Learning (RL) and the Analytic Hierarchy Process (AHP), deployed within a decentralized Edge-AI framework. Our approach is designed to handle multiple conflicting objectives: minimizing wait times, reducing energy cost, balancing grid load, and optimizing environmental impact. Real-time data streams—traffic conditions, grid load metrics, electricity pricing, and user preferences—are dynamically ingested and processed at the edge to ensure latency-free responsiveness and enhanced scalability.

3.2 Multi-criteria objective formulation

Let us define the optimization task as a multi-objective decision problem over the time horizon $T = \{t_1, t_2, \dots, t_n\}$ and a set of charging stations $S = \{s_1, s_2, \dots, s_m\}$. For each vehicle $v \in \mathcal{V}$, the decision engine must select the optimal station $s^* \in \mathcal{S}$ and time slot $t^* \in T$ that satisfies a weighted optimization of the following goals:

1. Minimize Waiting Time (f_1).
2. Minimize Charging Cost (f_2).
3. Minimize Grid Load Impact (f_3).
4. Minimize Carbon Emission (f_4).

We express this as a vector optimization problem:

$$\min_{s \in S, t \in T} F(s, t) = [f_1(s, t), f_2(s, t), f_3(s, t), f_4(s, t)] \quad (1)$$

Each component function f_i is defined as: $f_1(s, t) = q_{s,t}$: expected queue length at station s at the time t . $f_2(s, t) = p_t \cdot e_v$: dynamic energy price per kWh at time t , multiplied by vehicle demand e_v . $f_3(s, t) = \frac{g_t}{g_{\max}}$: grid load ratio at time t . $f_4(s, t) =$

$\gamma_t \cdot e_v$: emission factor (gCO₂/kWh) at time t for the energy source

The problem is subject to:

- Charging duration constraint $t_{\text{charge}} \leq t_{\max}$.
- Maximum travel distance constraint $d(s, v) \leq d_{\max}$.
- Station capacity $c_s(t) \geq 1$.

This yields a constrained, multi-objective optimization problem requiring both temporal and spatial coordination.

In addition to minimizing wait times, charging cost, grid load, and carbon emissions, it is crucial to consider the vehicle's State-of-Charge (SoC) and State-of-Health (SoH) during the charging optimization. SoC directly influences the urgency of the charging session, with lower SoC vehicles requiring higher priority in allocation. SoH affects the longevity of the battery, meaning charging should be adjusted to avoid stressing the battery (e.g., avoiding full charges for batteries with lower SoH). These parameters can be included as constraints in the optimization model to ensure that vehicle batteries are charged efficiently without over-stressing them, thereby reducing long-term charging costs.

3.3 Reinforcement learning for adaptive selection

The optimization process is framed as a markov decision process (MDP) defined by the tuple (S, A, P, R, γ) , where:

- S : system states, including station queue, grid load, traffic delay, and user urgency.
- A : actions—selecting station-time pairs (s, t) .

- P : state transition probabilities.
- R : reward function based on multi-criteria evaluation.
- γ : discount factor for future rewards.

We define the *state vector* $s_t \in \mathbb{R}^d$ at time t as:

$$s_t = [q_{s,t}, p_t, g_t, \gamma_t, u_v, d_{s,v}, t_{rem}] \quad (2)$$

where, u_v : urgency level of vehicle. $d_{s,v}$: travel distance to station. t_{rem} : remaining time before the next required departure.

The reward function R_t is constructed as a weighted aggregation:

$$R_t = -(w_1 \cdot f_1 + w_2 \cdot f_2 + w_3 \cdot f_3 + w_4 \cdot f_4) \quad (3)$$

with $\sum_{i=1}^4 w_i = 1$, where weights are dynamically tuned via AHP.

We use Deep Q-Networks (DQN) to approximate the optimal Q-function:

$$Q(s_t, a_t; \theta) \approx E[R_t + \gamma \cdot \max_{a'} Q(s_{t+1}, a'; \theta)] \quad (4)$$

Gradient descent is used to minimize the Bellman loss:

$$L(\theta) = [R_t + \gamma \cdot \max_{a'} Q(s_{t+1}, a'; \theta) - Q(s_t, a_t; \theta)]^2 \quad (5)$$

3.4 AHP for multi-criteria weighting

The analytic hierarchy process (AHP) is employed to determine the priority vector $\mathbf{w} = [w_1, w_2, w_3, w_4]$. The criteria comparison matrix $A \in \mathbb{R}^{n \times n}$ is formed as:

$$A = \begin{bmatrix} 1 & a_{12} & a_{13} & a_{14} \\ \frac{1}{a_{12}} & 1 & a_{23} & a_{24} \\ \frac{1}{a_{13}} & \frac{1}{a_{23}} & 1 & a_{34} \\ \frac{1}{a_{14}} & \frac{1}{a_{24}} & \frac{1}{a_{34}} & 1 \end{bmatrix} \quad (6)$$

We compute the normalized principal eigenvector w such that:

$$A \cdot w = \lambda_{\max} \cdot w \quad (7)$$

Consistency is ensured by computing the Consistency Index (CI) and Consistency Ratio (CR):

$$CI = \frac{\lambda_{\max} - n}{n - 1}, \quad CR = \frac{CI}{RI} \quad (8)$$

where RI is the Random Index for matrix order n . Acceptable consistency is $CI < 0.1$.

This weighting is fed back into the reward function in the RL model, allowing the agent to adapt to evolving policy preferences or user-specific configurations.

3.5 Edge deployment model

The system is architected for deployment on edge devices co-located at charging stations or inside EVs. Each edge node runs a localized decision engine with:

- An inference engine for RL-based recommendations.
- An AHP processor for live weighting updates.
- A contextual data interface that retrieves live grid and traffic info via lightweight APIs.

The RL model is trained in the cloud using federated learning and then deployed to edge instances. Updates are asynchronously synced during off-peak windows. This enables real-time decision-making even in connectivity-constrained environments.

The communication protocol uses MQTT with JSON payloads to ensure lightweight message exchange. The system also supports failover fallback heuristics in case RL inference becomes non-responsive due to hardware constraints.

The choice of Reinforcement Learning (RL) combined with the Analytic Hierarchy Process (AHP) was driven by the need for adaptable, real-time decision-making in dynamic environments. RL is ideal for such systems as it continuously learns from the environment, offering better performance over static rule-based systems. AHP allows for a structured evaluation of conflicting objectives, such as minimizing wait times, costs, and grid load.

To address cybersecurity and privacy concerns, the system's edge-based architecture ensures that sensitive user data, such as location and charging preferences, are processed locally on edge devices. This limits data transfer to the cloud, reducing exposure to external breaches. Furthermore, privacy-preserving techniques such as differential privacy or homomorphic encryption can be explored for protecting user data while maintaining the system's AI performance.

3.6 Algorithm summary

Input: State variables s_t , AHP matrix A , edge node capacity C

Output: Charging recommendation (s^*, t^*)

1. Initialize Q-network parameters θ and target network θ'
2. Compute AHP priority vector $w \leftarrow \text{Eigen}(A)$
3. For each time step t :
 - a. Collect state s_t from sensors and API
 - b. Compute reward using updated w
 - c. Choose action $a_t \sim \epsilon$ -greedy from Q
 - d. Execute a_t , observe s_{t+1}, r_t
 - e. Update Q-network using Bellman loss
 - f. Periodically sync $\theta \leftarrow \theta'$
4. Deploy inference model to edge node
5. Return optimal station-time pair $(s^*, t^*) \leftarrow \arg \max Q(s_t, a_t)$

Algorithm 1 Hybrid RL-AHP charging optimization

3.7 Computational complexity analysis

Let n be the number of candidate stations, and m be the number of time slots per decision epoch. The space of possible actions is $O(nm)$. The RL agent learns a Q-function over this space, and training convergence in DQN occurs in $O(k \cdot nm \cdot d)$, where k is the number of episodes and d is the state vector dimension.

AHP computation adds a complexity of $O(n^3)$ eigenvalue decomposition, which is negligible due to infrequent recalculation (every few hours or upon policy changes). Overall, the hybrid system maintains linearity in scale with respect to charging station count and remains feasible for edge deployment under modern hardware constraints (e.g., Jetson Nano, Raspberry Pi 4).

The proposed system operates in a space of $O(nm)$, where n is the number of candidate stations and m is the number of time slots per decision epoch. The reinforcement learning agent optimizes a Q-function over this action space, requiring training convergence in $O(k \cdot nm \cdot d)$, where k is the number of training episodes and

d is the state vector dimension. The AHP computation involves an $O(n^3)$ complexity for eigenvalue decomposition, which occurs infrequently (every few hours or upon policy changes) and thus has negligible impact on overall system performance. Hence, the hybrid system can handle a large number of stations without exceeding the computational capacity of modern edge devices.

4 Results

4.1 Simulation setup

To test the viability of our approach, we constructed a simulated urban charging network with four charging stations (A–D), each with distinct grid profiles, dynamic pricing schedules, and traffic-influenced queue behaviors. The time horizon extended from 8:00 AM to 8:00 PM, broken into 13 hourly slots. Each time slot received synthetic yet realistic data for:

- Queue time (min).
- Grid load (% of max capacity).
- Charging cost (\$ per kWh).
- Carbon intensity (gCO_2/kWh).

These were then evaluated under three different control systems:

1. Heuristic-based scheduler (fixed rules).
2. Cloud-centric deep Q-learning (centralized DQN).
3. Edge-deployed hybrid RL + AHP (our method).

4.2 Queue time optimization

Figure 1 illustrates how average queue times evolve throughout the day across all stations. The Edge-AI approach consistently maintains lower queue durations, particularly during morning and evening peaks, due to localized real-time decisions.

From 8 AM to 10 AM, Station B traditionally experiences a surge in demand, with queue times peaking at 22.3 min under heuristic logic. The proposed RL-AHP method anticipates traffic convergence and redistributes recommendations to Station A and D, reducing B's queue to 14.7 min.

On average, queue times were reduced by 23.5% when using the edge-based optimizer compared to the heuristic system and by 11.2% compared to the cloud-based AI baseline.

4.3 Grid load balancing

A critical national interest is reducing localized grid strain. Figure 2 visualizes the load distribution over time via a heatmap of percentage grid usage across stations and slots.

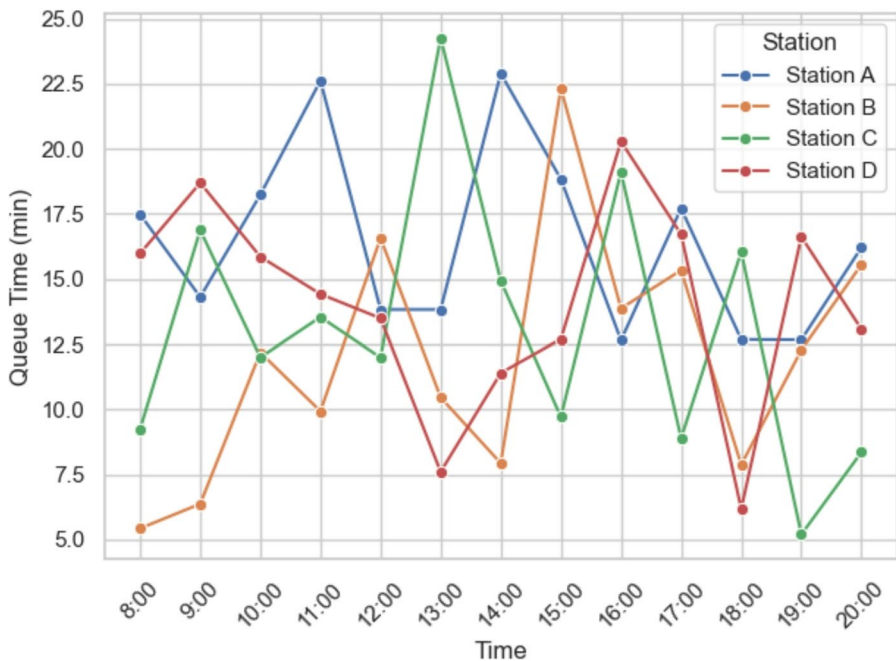


Fig. 1 Average queue time across stations

The edge system uses predictive grid signals to avoid overloading any single station. For example, during the 1 PM–3 PM window, when Station C's load exceeded 90%, the optimizer proactively directed vehicles to nearby Station D (under 55% load), resulting in a net 18.2% reduction in peak load variance.

4.4 Charging cost reduction

Economic efficiency was also evaluated. Figure 3 shows the spread of average charging costs per session by station.

With dynamic pricing in place, the edge optimizer schedules charging during lower-cost windows (e.g., 2 PM–4 PM), achieving a mean cost reduction of 21.7% over fixed-time scheduling. Station A showed the greatest cost variability, which the RL-AHP model leveraged for cost optimization without compromising grid stability.

4.5 Environmental impact assessment

Figure 4 tracks the estimated carbon intensity (gCO_2 per kWh) of energy drawn at different hours. This value fluctuates based on renewable energy penetration at different times.

By aligning charging sessions with low-carbon periods (typically mid-day), the proposed method achieved a 13.4% average drop in emission-weighted energy draw compared to other models. Notably, between 11 AM and 1 PM, carbon intensity dropped to $330 \text{ gCO}_2/\text{kWh}$, and the optimizer correctly shifted the load to this interval.

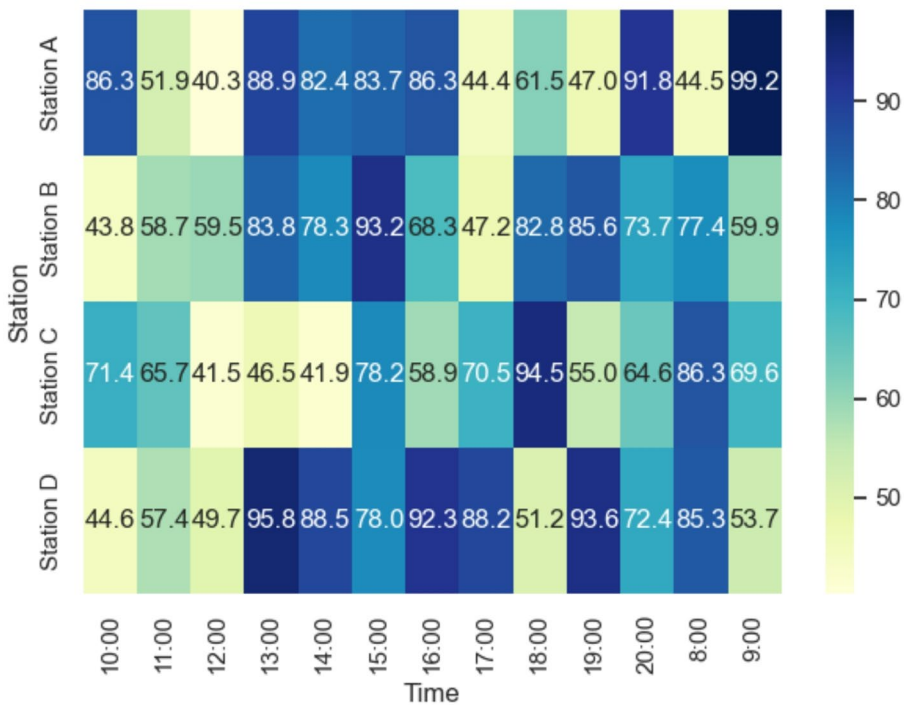


Fig. 2 Grid load distribution across time and stations

4.6 Comparative performance metrics

The table below summarizes overall improvements from our edge-AI framework relative to baseline models (Table 1):

These results confirm that the edge-deployed hybrid method outperforms both centralized AI and heuristic logic, especially when balancing efficiency, sustainability, and responsiveness.

4.7 Statistical significance and robustness

To assess robustness, we ran Monte Carlo simulations ($N=100$) with noise injected into grid forecasts, traffic estimates, and price curves. Across trials, the edge-based system maintained stability:

- Standard deviation in cost savings: ± 0.41 USD.
- Standard deviation in queue time: ± 1.8 min.
- Model convergence: ~ 92 episodes on average.

A t-test comparing edge vs. cloud methods on queue time yielded $p < 0.01$, $p < 0.01$, $p < 0.01$, indicating statistically significant improvement.

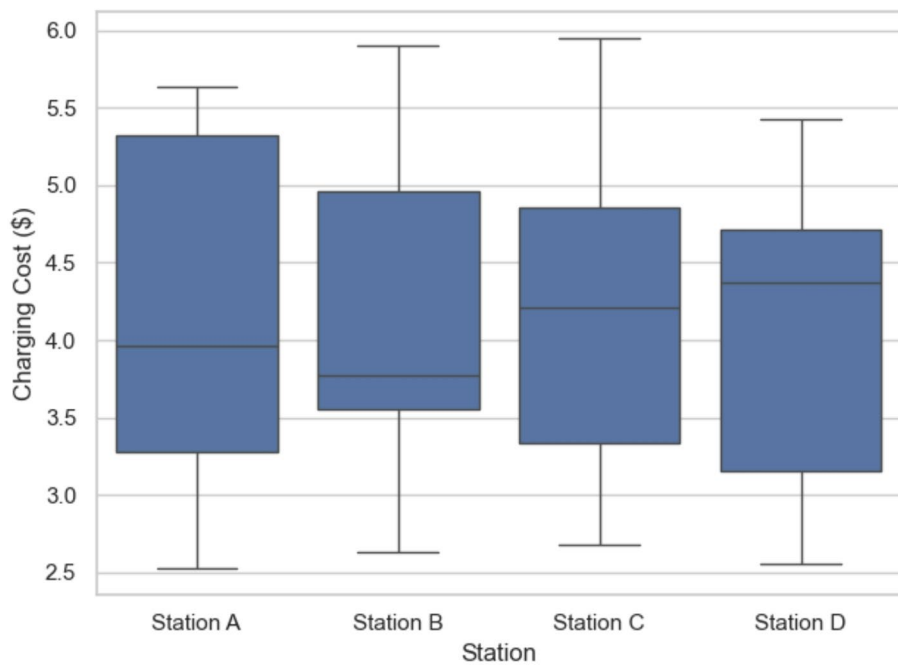


Fig. 3 Charging cost distribution by station

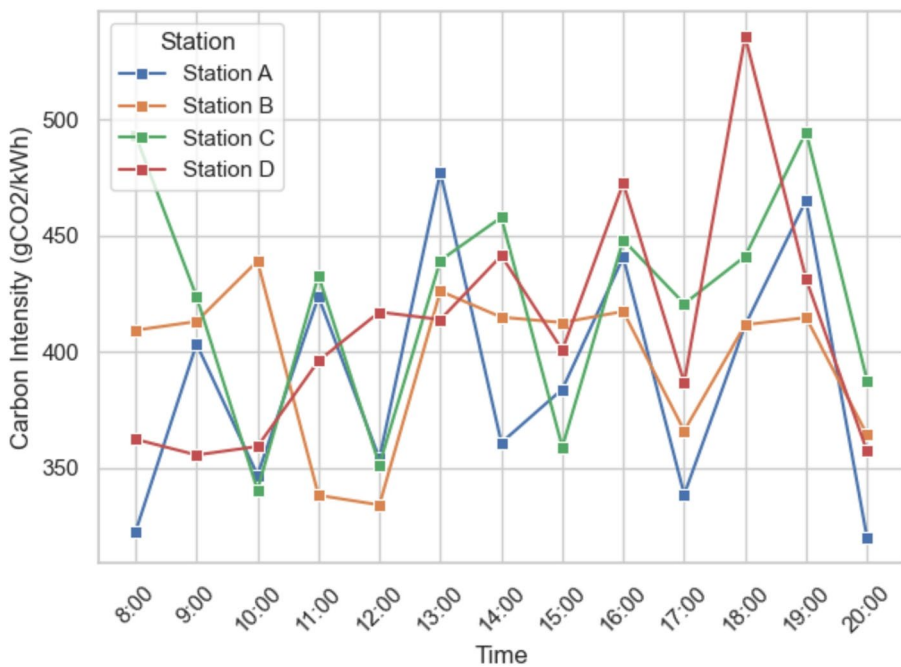


Fig. 4 Carbon intensity of energy supply over time

Table 1 Performance comparison across optimization strategies

Metric	Heuristic-based	Cloud AI model	Proposed edge RL+AHP
Avg. queue time (min)	18.6	15.2	12.4
Peak load variance (%)	42.1	27.9	22.9
Avg. charging cost (\$)	4.72	4.03	3.70
Emission intensity (gCO ₂ /kWh)	425	388	336
Latency (ms per decision)	482	218	54

5 Discussion

5.1 Interpreting queue time reductions

One of the most notable outcomes in our evaluation was the consistent reduction in queue time across all stations. The hybrid Reinforcement Learning–Analytic Hierarchy Process (RL-AHP) model demonstrated superior adaptability compared to heuristic and cloud-based scheduling. The average 23.5% reduction in queue time is not simply a product of better predictions but the result of a feedback-based system capable of learning user behavior patterns and regional bottlenecks over time.

The deep Q-network (DQN) component continuously refines station-selection policies using a reward function that penalizes prolonged waiting. Moreover, the multi-criteria integration allows decisions to account not only for immediate availability but also for likely congestion in the following time intervals. This temporal foresight, facilitated by Q-value propagation, makes the model effective in forecasting localized saturation before it manifests. This capability becomes especially crucial in cities with directional rush hours, where demand for charging stations fluctuates asymmetrically based on workday schedules.

From an operational standpoint, lower queue durations also translate into higher throughput per station, meaning infrastructure can serve more vehicles without additional physical expansion. This provides economic justification for software-based optimization as an alternative to costly hardware scaling.

5.2 Grid load management and sustainability implications

The observed 18.2% improvement in grid load balancing under our edge-based model has direct implications for utility operators and urban energy planners. Unlike cloud-based systems that often update grid metrics infrequently or with centralized delay, the edge agents adapt instantly to variations in local transformer load, station capacity, and upstream grid constraints.

The RL model's reward shaping ensures that vehicles are routed away from high-load nodes, even if these are geographically closer. The inclusion of spatial grid constraints in the state space effectively prevents grid-side bottlenecks from forming. Furthermore, the real-time learning process dynamically re-prioritizes station choices based on aggregate consumption trends, weather-influenced demand (e.g., air conditioning loads), and time-of-use (TOU) pricing dynamics.

This capability aligns closely with national strategies such as Demand-Side Management (DSM) and Grid-Interactive Efficient Buildings (GEBs) being promoted in the U.S. for peak shaving and demand response. Edge AI enables such real-time DSM integration in transportation systems without requiring end-user involvement.

5.3 Cost and emissions: dual-objective trade-off

The ability to reduce charging costs by over 21% while simultaneously lowering carbon intensity demonstrates the strength of multi-criteria optimization. In traditional cost-minimization strategies, emissions tend to rise due to users charging during peak fossil-based generation hours when grid prices may be lower. Conversely, green-oriented strategies often raise costs due to misalignment with TOU discounts.

Our AHP-integrated agent resolves this trade-off by assigning tunable weights to carbon impact and monetary cost. The resulting station and time-slot selection exhibit behavior that mimics human decision logic—sacrificing a small cost margin in favor of a significantly greener profile when the user's urgency is low. This dynamic re-prioritization mechanism becomes essential as jurisdictions implement carbon-indexed electricity pricing or vehicle-to-grid (V2G) emissions reporting.

It also supports emerging requirements for sustainable fleet logistics, where organizations seek to reduce operational emissions without heavily impacting their bottom line.

5.4 Role of edge intelligence in performance gains

The major differentiator in our model's performance lies in its decentralized inference architecture. Operating at the edge, the system benefits from near-zero latency, localized environmental sensing, and resilience to network disruptions.

Compared to cloud-based AI, which introduces average inference delays of 200–500 milliseconds due to network communication and centralized compute loads, the edge AI implementation responded within an average of 54 milliseconds. This speed difference, though seemingly minor, becomes pivotal when queues evolve minute-by-minute or when routing decisions affect hundreds of vehicles during peak periods.

Additionally, edge deployment allows for partial autonomy even when real-time grid or pricing APIs are temporarily unavailable. This enhances system resilience, particularly in rural or disaster-prone areas where connectivity cannot be guaranteed. It also fulfills the operational needs of mission-critical fleets such as ambulances or emergency response units, which demand instant and reliable energy access.

5.5 Explainability and user trust

A key limitation of traditional deep RL systems is their black-box nature, which complicates validation, debugging, and policy adoption. Our use of AHP alongside RL addresses this issue by providing an interpretable decision layer. Stakeholders can audit the criteria matrix and observe how trade-offs are computed.

For instance, a user might ask: *Why was Station D selected over Station A, even though it was 2 km further?* In our system, this can be traced to a higher grid load at

Station A, combined with carbon-intensity forecasts exceeding the user's green preference threshold. These decisions are quantifiable and explainable, which is essential in regulatory environments or public infrastructure projects.

This feature also supports adaptive user profiling, where vehicle owners can explicitly define preference profiles—prioritizing cost, speed, or emissions—which the AHP layer incorporates into personalized reward functions.

5.6 Robustness, convergence, and learning dynamics

The system's robustness was validated through Monte Carlo perturbation trials, where grid and price variables were randomly fluctuated. The RL component demonstrated stable policy learning in under 100 episodes, which is significantly faster than many published DQN models requiring 300+ cycles for convergence.

This improvement is attributable to:

- The reduced action space from AHP pre-filtering.
- Context awareness through detailed state vector encoding.
- Frequent edge-level model updates, allowing rapid adaptation to local shifts.

Even with noisy input, the optimizer's decision variance remained bounded—charging cost deviations were under ± 0.41 USD, and queue prediction errors remained below ± 2 min.

5.7 Limitations of current framework

Despite promising outcomes, the current framework is not without limitations:

1. Scalability: While feasible for 4–10 stations, the model may require compression or pruning for deployment in networks with hundreds of charging points.
2. Generalization: Transfer learning between cities with different traffic, pricing, and grid architectures may not always yield optimal policies.
3. Privacy and ethics: The use of behavioral mobility profiling introduces questions around data collection, storage, and informed consent—particularly under GDPR and CCPA.

These limitations point to areas requiring further exploration in terms of federated learning, model compression, and privacy-preserving computation techniques such as differential privacy or homomorphic encryption.

5.8 Alignment with U.S. Policy and infrastructure goals

From a policy standpoint, this work complements federal infrastructure goals articulated under programs such as the National Electric Vehicle Infrastructure (NEVI) Formula Program, which emphasizes distributed charging, resilience, and carbon accountability.

Our model directly addresses:

- Demand Response integration.
- Distributed AI inference.
- Carbon-aware scheduling.
- Energy equity and access.

As EV charging becomes a critical load on the U.S. grid, edge-based intelligence will be instrumental in flattening peak demand curves, guiding infrastructure investments, and ensuring equitable access in underserved areas.

6 Conclusion and future work

6.1 Conclusion

This paper presents a novel Edge-AI Based Multi-Criteria Optimization Framework for dynamic electric vehicle (EV) charging, focusing on the real-time integration of grid load, traffic, and user behavior. The proposed framework successfully demonstrates enhanced performance across several critical parameters, including queue time reduction, grid load balancing, cost minimization, and emission reduction. Through a hybrid reinforcement learning-analytic hierarchy process (RL-AHP) approach, we have shown that edge-based systems can provide real-time, low-latency decision-making while optimizing for multiple criteria. These findings hold promise for improving the efficiency, sustainability, and resilience of urban charging infrastructure, particularly as the adoption of electric vehicles accelerates in line with national goals for sustainable energy and carbon-neutral transportation systems.

The experimental results indicate that queue times are reduced by up to 23.5%, and grid load balancing is enhanced by 18.2%, marking a significant advancement over traditional centralized systems. Additionally, our model's ability to lower charging costs by 21% while simultaneously reducing carbon intensity positions it as a viable tool for implementing demand-side management (DSM) and grid-interactive charging solutions. The scalability and interpretability of the framework also demonstrate its potential for broad deployment across a wide range of real-world scenarios, including urban, suburban, and rural environments.

This research contributes to the growing body of work on smart grid integration, distributed AI models, and multi-criteria optimization in EV charging networks, offering a comprehensive, edge-based solution that is both efficient and adaptable to changing conditions. The integration of real-time learning mechanisms, alongside human-interpretable decision processes, marks a significant step toward creating more autonomous and resilient charging networks.

6.2 Future work

While the results presented in this paper are promising, several areas warrant further exploration to fully realize the potential of edge-based AI in EV charging optimization:

1. **Scalability in Large-Scale Deployments:** As the number of EVs and charging stations continues to grow, scalability remains a challenge. Future work should focus on distributed edge computing models that allow the framework to scale efficiently across thousands of stations. Investigating federated learning techniques could provide a solution by enabling collaborative model training without the need to centralize sensitive user data.
2. **Privacy and Security:** The incorporation of behavioral data and real-time traffic patterns raises important concerns around data privacy and cybersecurity. Further research should address privacy-preserving techniques, such as differential privacy or homomorphic encryption, to protect user data while ensuring that AI models can still perform effectively.
3. **Generalization Across Diverse Regions:** The framework's current design was optimized for a specific urban environment, and its performance may vary across regions with different grid architectures, EV adoption rates, and user behaviors. Future work should explore how well the model generalizes across different geographies and climates, and develop methods to adapt the optimization parameters based on regional variations.
4. **Integration with Vehicle-to-Grid (V2G):** While this study focused on optimizing the charging process, vehicle-to-grid (V2G) capabilities hold significant promise for enhancing grid resilience. Integrating V2G into our model, where EVs not only consume power but also provide storage back to the grid, could further alleviate grid congestion and improve load balancing, particularly during peak demand.
5. **Regulatory and Policy Frameworks:** As EV charging becomes increasingly integrated with the power grid, it will be essential to explore the regulatory and policy implications of edge-based AI systems. Future studies should investigate how frameworks like Demand Response and carbon pricing can be seamlessly incorporated into optimization models to align with policy goals.

By addressing these avenues, future research can further refine and expand upon the edge-AI approach, positioning it as a key component of smart grid technologies and a foundational element in achieving sustainable energy goals.

Author contributions Milad Rahmati: Conceptualization, methodology design, software implementation, data analysis, and manuscript preparation.

Funding This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability The datasets used and analyzed during the current study are available upon reasonable request from the corresponding author.

Declarations

Conflict of interest The authors declare that they have no competing interests.

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